

Uncertainty-Aware AI Prioritization of Boron Nitride Materials for UV and 2D Dielectric Applications

Research Plan

Principal investigator, position and affiliated department

Prof. Ma Chen, Assistant Professor, Department of Computer Science, City University of Hong Kong.

Collaborator(s) and respective department(s), if any

Shenzhen Zhongneng Tianyi Technology Co., Ltd. (industrial collaborator).

Duration of the project

Abstract

Boron nitride (BN) materials are promising for wide-band-gap ultraviolet optoelectronics and dielectric support layers in two-dimensional heterostructures. This project aims to build an uncertainty-aware AI pipeline to prioritize BN-related materials for these two application targets. The study will construct a bounded BN design space, learn property and uncertainty models from public materials databases, rank candidates under band-gap, stability, novelty, and domain-distance constraints, and generate first-pass structure hypotheses for selected candidates. Expected outcomes include a curated BN-centered dataset, benchmarked property models, an uncertainty-aware ranked candidate table, structure handoff files, and manuscript-ready analysis for academic and industrial collaboration.

Key objectives

1. Define a bounded BN design space.

$$\mathcal{C}_{\text{BN}} = \mathcal{C}_{\text{core}} \cup \mathcal{C}_{\text{ext}} \cup \mathcal{C}_{\text{explore}},$$

where $\mathcal{C}_{\text{core}} = \{\text{BN}, \text{h-BN}, \text{c-BN}, \text{doped BN}\}$, $\mathcal{C}_{\text{ext}} = \{\text{BCN}, \text{BC}_2\text{N}, \text{SiBN}\}$, and $\mathcal{C}_{\text{explore}}$ is restricted by charge neutrality, small-cell prototypes, and BN-family structural motifs:

$$\mathcal{C}_{\text{explore}} = \{c : q(c) = 0, N_{\text{atom}}(c) \leq 12, \pi(c) \in \{\text{BN}, \text{BCN}, \text{BC}_2\text{N}, \text{SiBN}\}\}.$$

Substitutions will be considered only when they can be mapped to B, N, C, or Si sites in these motifs. *Output:* a curated dataset with formula, structure, target-property, and provenance fields.

2. Learn property and uncertainty models.

$$f_{\theta} : (x_{\text{form}}, x_{\text{struc}}) \rightarrow \mathbf{y} = (E_{\text{g}}, \Delta H, E_{\text{hull}}, \epsilon_r, \rho_{2\text{D}}, u).$$

Output: benchmarked models for band gap, stability proxy, dielectric relevance, and predictive uncertainty.

3. Prioritize candidates under application constraints.

$$U(c) = w_1 \rho_{\text{gap}}(c) - w_2 E_{\text{hull}}(c) - w_3 \hat{u}(c) - w_4 d_{\text{dom}}(c) + w_5 \rho_{\text{novel}}(c) + w_6 \rho_{\text{app}}(c).$$

Output: a ranked table of candidate families with predicted properties, uncertainty, novelty score, and action labels.

4. Generate structure hypotheses for selected candidates.

$$c \mapsto \{s_{c,1}, \dots, s_{c,k}\} \mapsto \{\text{geometry}, \text{relaxation}, \text{property recheck}\}.$$

Output: structure files and a validation handoff package for high-priority candidates.

Research methodology

WP1. Data construction and cleaning

The project starts from public computational materials resources rather than an unbounded chemical search. 2DMPedia provides more than 6,000 monolayer structures generated by top-down and bottom-up strategies [1]. JARVIS-Tools supports programmatic access to atomistic materials datasets [2], while Matbench, matminer, the Materials Project, and C2DB provide benchmark, featurization, and reference spaces for materials ML [3–6]. Records will be deduplicated, aligned to target-property definitions, and labelled with provenance:

$$r_i = (c_i, s_i, \mathbf{y}_i, p_i), \quad \mathcal{D}_{\text{BN}} = \{r_i : c_i \in \mathcal{C}_{\text{BN}}\}, \quad p_i = (\text{source}, \text{method}, \text{version}).$$

This creates the BN-centered data layer that supports all later modelling and ranking decisions.

WP2. Property prediction and uncertainty estimation

The modelling target is a decision vector rather than a single benchmark number:

$$\max_{c,s} \mathbf{g}(c,s) = [\rho_{\text{gap}}, -E_{\text{hull}}, \rho_{\text{dielectric}}, \rho_{\text{UV}}, \rho_{\text{2D}}, -u].$$

The band-gap term will be treated as a target-window score rather than a monotonic objective, because UV optoelectronic relevance depends on the desired spectral range and the direct/indirect nature of the gap:

$$\rho_{\text{gap}}(c) = \exp \left[-\frac{(\hat{E}_{\text{g}}(c) - E_{\text{g,app}}^*)^2}{2\sigma_{\text{app}}^2} \right] \cdot \eta_{\text{direct}}(c).$$

The application scores $\rho_{\text{dielectric}}$, ρ_{UV} , and ρ_{2D} are proxy scores rather than direct experimental labels. They will be derived from available computed properties, dimensionality metadata, and rule-based target criteria. Model families will include descriptor baselines, composition neural networks, and crystal graph models, drawing on established materials-ML tools such as pymatgen, CGCNN, MEGNet, ALIGNN, CrabNet, and CHGNet [7–12]. Predictive uncertainty will be estimated using model ensembles and calibrated against validation-set errors:

$$\hat{u}_{\text{cal}}(c) = \alpha \cdot \text{Std}_{m \in \mathcal{M}}[f_m(c)] + \beta \cdot \widehat{\text{err}}_{\text{val}}(c).$$

Candidates with high predicted performance but high uncertainty will be labelled as exploratory rather than priority candidates. These models are not the novelty by themselves; they provide a controlled basis for a BN-specific decision pipeline.

Formula-only candidates will first be screened using composition descriptors and composition-based models. Candidates passing the first screen will then enter structure hypothesis generation, after which structure-resolved graph models and validation checks will be applied. If structure-resolved models do not outperform composition or descriptor baselines under BN-family-aware splits, the ranking layer will prioritize robust baseline predictions and mark graph-model outputs as exploratory evidence only.

WP3. Multi-objective ranking and candidate selection

The output for each candidate will be a decision vector

$$\mathbf{z}(c) = (\hat{E}_{\text{g}}, \rho_{\text{gap}}, E_{\text{hull}}, \hat{e}_r, \hat{u}, d_{\text{dom}}, \rho_{\text{novel}}, \rho_{\text{app}}, a),$$

where domain distance will be measured using composition descriptors, structural descriptors, and learned embedding distances from the BN-centered training set:

$$d_{\text{dom}}(c) = \min_{r_i \in \mathcal{D}_{\text{BN}}} \|\phi(c) - \phi(c_i, s_i)\|_{\Sigma}.$$

Novelty will be estimated by distance from known BN-family compounds, while application relevance will be assigned according to whether the candidate matches the wide-gap/UV or dielectric/2D-support target. The action label is

$$a \in \{\text{control}, \text{priority}, \text{explore}, \text{hold}\}.$$

Candidates will be prioritized only when predicted performance, uncertainty, and domain-distance criteria are jointly satisfied:

$$a(c) = \text{priority} \quad \text{if} \quad U(c) \geq q_{\alpha}, \quad \hat{u}(c) \leq u_{\text{max}}, \quad d_{\text{dom}}(c) \leq d_{\text{max}}.$$

This connects prediction to research action: reference controls, high-priority validation, exploratory families, and candidates reserved for later evidence.

The candidate selection protocol is: generate a BN-related candidate pool; predict target properties; estimate uncertainty and domain distance; filter by stability, uncertainty, and application target; assign action labels. Priority candidates proceed to structure handoff and validation; exploratory candidates are retained as high-risk families; control candidates serve as reference checks; hold candidates are deferred until additional evidence is available.

WP4. Structure generation, validation handoff, and iteration

Selected formulas will first be converted into candidate structures through prototype reuse, database-derived structure transfer, and controlled substitution. Generative models such as CDVAE will be used only when prototype-based routes are insufficient [13]. A first-pass handoff package will include structure files, predicted properties, uncertainty scores, provenance, and recommended validation actions.

The validation protocol will test both interpolation and extrapolation. Models will be evaluated using random splits, grouped-by-formula splits, and BN-family-aware splits. Candidate ranking will be assessed by model ensembles and perturbation of objective weights:

$$\text{MAE}, \text{RMSE}, R^2, \tau_{\text{Kendall}}, \rho_{\text{Spearman}}, P(c \in \text{Top-}k).$$

For selected priority candidates, materials-side validation will include charge-neutrality checks, structural sanity checks, geometry relaxation where feasible, energy-above-hull re-estimation,

$$E_{\text{hull}}(c, s) = E(c, s) - E_{\text{hull}}(\text{composition}(c)),$$

and comparison with known BN-family references. Stability will be handled separately for bulk-like and 2D candidates: bulk-like entries will use energy-above-hull or formation-energy proxies, while 2D entries will additionally use dimensionality, database provenance, and available 2D stability indicators. The current local project already provides a starting implementation for public-data ingestion, BN-themed candidate generation, model comparison, rank stability, and first-pass structure handoff. The proposed research will extend this base into a larger, uncertainty-aware BN prioritization platform.

Significance and major deliverables

BN is a high-value family for wide-band-gap ultraviolet materials, dielectric layers, and 2D heterostructure platforms^[14–16]. This proposal focuses on two targets only: UV/wide-band-gap candidates and dielectric/2D-support candidates. The scientific question is:

Which BN-related compositions and structures should be investigated next, and with what level of uncertainty?

The novelty lies in the BN-centered data layer, uncertainty-aware candidate selection, multi-objective ranking under application constraints, formula-to-structure handoff workflow, and rank-stability analysis for materials prioritization.

Major deliverables

1. **D1:** BN-centered dataset from at least three public sources, with formula, structure, target-property, dimensionality, and provenance fields.
2. **D2:** Benchmark report covering random, grouped-by-formula, and BN-family-aware splits with MAE, RMSE, R^2 , calibration, and rank-stability metrics.
3. **D3:** Ranked table of at least 30 candidate families, including formula, predicted band gap, stability proxy, uncertainty, domain distance, novelty score, application label, and recommended action; 5–10 high-priority candidates will be selected for structure-level handoff.
4. **D4:** Structure hypothesis package for selected candidates, including CIF/POSCAR files, provenance record, predicted properties, uncertainty score, and validation notes.
5. **D5:** Technical report, demo, manuscript-ready figures, and a concise candidate-prioritization summary for the PI and collaborator.

Ethics and safety

The initial project is computational and data-driven. It uses public databases, software, models, and generated candidate structures. No human subjects, living animals, wet-lab synthesis, biological samples, or radiation work will be conducted in the initial computational phase. If later work involves chemical synthesis or regulated laboratory procedures, the responsible experimental party will handle the required safety approvals.

Selected references

1. Zhou, J. et al. 2DMatPedia: an open computational database of two-dimensional materials from top-down and bottom-up approaches. *Scientific Data* 6, 86 (2019).
2. Choudhary, K. et al. JARVIS-Tools: open-access software for atomistic data-driven materials design. *npj Computational Materials* 6, 173 (2020).
3. Dunn, A. et al. Benchmarking materials property prediction methods: the Matbench test set and Automatminer reference algorithm. *npj Computational Materials* 6, 138 (2020).
4. Ward, L. et al. Matminer: An open source toolkit for materials data mining. *Computational Materials Science* 152, 60–69 (2018).
5. Jain, A. et al. Commentary: The Materials Project: a materials genome approach to accelerating materials innovation. *APL Materials* 1, 011002 (2013).
6. Hastrup, S. et al. The Computational 2D Materials Database: high-throughput modeling and discovery of atomically thin crystals. *2D Materials* 5, 042002 (2018).
7. Ong, S. P. et al. Python Materials Genomics (pymatgen): A robust, open-source python library for materials analysis. *Computational Materials Science* 68, 314–319 (2013).
8. Xie, T. and Grossman, J. C. Crystal graph convolutional neural networks for an accurate and interpretable prediction of material properties. *Physical Review Letters* 120, 145301 (2018).

9. Chen, C. et al. Graph networks as a universal machine learning framework for molecules and crystals. *Chemistry of Materials* 31, 3564–3572 (2019).
10. Choudhary, K. and DeCost, B. Atomistic line graph neural network for improved materials property predictions. *npj Computational Materials* 7, 185 (2021).
11. Wang, A. Y.-T. et al. CrabNet for explainable deep learning in materials science. *npj Computational Materials* 7, 77 (2021).
12. Deng, B. et al. CHGNet as a pretrained universal neural network potential for charge-informed atomistic modelling. *Nature Machine Intelligence* 5, 1031–1041 (2023).
13. Xie, T. et al. Crystal diffusion variational autoencoder for periodic material generation. *ICLR* (2022).
14. Watanabe, K., Taniguchi, T. and Kanda, H. Direct-bandgap properties and evidence for ultraviolet lasing of hexagonal boron nitride single crystal. *Nature Materials* 3, 404–409 (2004).
15. Cassabois, G., Valvin, P. and Gil, B. Hexagonal boron nitride is an indirect bandgap semiconductor. *Nature Photonics* 10, 262–266 (2016).
16. Dean, C. R. et al. Boron nitride substrates for high-quality graphene electronics. *Nature Nanotechnology* 5, 722–726 (2010).

Project Details

Project Title: Uncertainty-Aware AI Prioritization of Boron Nitride Materials for UV and 2D Dielectric Applications

PI (Dept): Prof. Ma Chen (Department of Computer Science)

Budget Breakdown

Budget Item	Detail breakdown	Amount (HKD)
Staffing	Student helper / research assistant for data cleaning, benchmark scripts, figure generation, documentation, and demo maintenance.	
Equipment	GPU cloud hours for graph-model training and ensemble inference; storage for downloaded structures, generated CIF/POSCAR files, benchmark logs, and model checkpoints.	
General expenses	Cloud credits, repository backup, software services, lightweight hosting for the demo, and data-management costs.	
Conference	Presentation materials and registration for AI-for-science, materials informatics, or applied computing venues.	
Overseas Travel	Optional PI-approved collaborator visit, conference travel, or partner technical meeting for candidate-review discussion.	
Total Funding		